

British Math Olympiad 2013 Round 1 — P3/6

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A number written in base 10 is a string of 3^{2013} digit 3s. No other digit appears. Find the highest power of 3 which divides this number.

Solution

Let $N = 3 \cdot 10^{3^{2013}-1} + 3 \cdot 10^{3^{2013}-2} + \dots + 3 \cdot 10^1 + 3 \cdot 10^0 = 3 \cdot \frac{10^{3^{2013}} - 1}{10 - 1}$. We claim that $v_3(N) = 3^{2013}$. It suffices to show that $v_3\left(\frac{1}{3}(10^{3^{2013}} - 1)\right) = 3^{2013} \iff v_3(10^{3^{2013}} - 1) = 3^{2013} + 1$.

First we show that $10^{3^{2013}} \equiv 1 \pmod{3^{3^{2013}+1}}$. Note that a positive integer is coprime with $3^{3^{2013}+1}$ as long as it doesn't have a factor of 3. We know that every third number of the positive integers is a multiple of 3, and thus the number of positive integers that are coprime with $3^{3^{2013}+1}$ is exactly $\frac{2}{3} \cdot 3^{3^{2013}+1} = 2 \cdot 3^{3^{2013}}$ and thus $\varphi(3^{3^{2013}+1}) = 2 \cdot 3^{3^{2013}}$. Now notice that $1 \equiv 10^{2 \cdot 3^{2013}} \equiv 10^{3^{2013}} \pmod{3^{3^{2013}+1}}$ by Euler's theorem, and this establishes that $3^{3^{2013}+1} \mid 10^{3^{2013}} - 1$ as desired.

Now we just need to show that $3^{3^{2013}+2} \nmid 10^{3^{2013}} - 1$. By the same argument as last time we know that $\varphi(3^{3^{2013}+2}) = \frac{2}{3} \cdot 3^{3^{2013}+2} = 2 \cdot 3^{3^{2013}+1}$. Assume, for the sake of contradiction, that $3^{3^{2013}+2} \mid 10^{3^{2013}} - 1$. Thus $10^{3^{2013}} - 1 \equiv 3^{2 \cdot 3^{2013}} - 1 \equiv 3^{(3^{2013}+2)+(3^{2013}-2)} - 1 \equiv -1 \equiv 0 \pmod{3^{3^{2013}+2}}$ which is a contradiction. Thus $3^{3^{2013}+2} \nmid 10^{3^{2013}} - 1$ and so $v_3(10^{3^{2013}} - 1) = 3^{2013} + 1$ as desired. Thus the highest power of 3 dividing N is 3^{2013} . ■